From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE (Advanced), 2014

FULL TEST – II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 246**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-A (07 – 10)** contains 4 multiple choice questions having one or more than one correct answer and each question carries **+4 marks** for correct answer and there is no negative marking.

(iii) **Section-A (11 – 14)** contains 4 Assertion-Reasoning (multiple choice questions) which have only one correct answer. Each question carries **+3 marks** each for correct answer and **– 1 mark** for wrong answer.

(iv) **Section-A (15 – 23)** contains 3 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

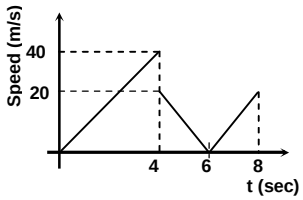
PART – I

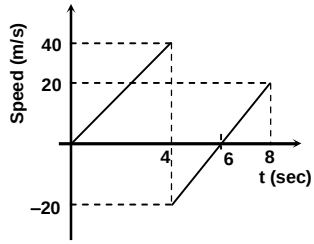
SECTION – A

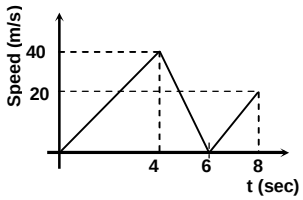
Straight Objective Type

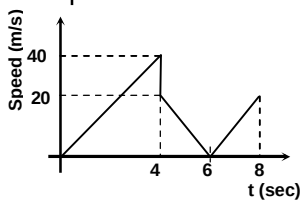
This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. A ball is dropped from a height of 80 m on a floor. At each collision, the ball losses half of its speed. Then the graph between speed versus time is

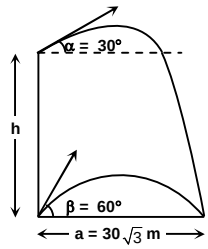
(A)


(B)


(C)


(D)


2. Shots are fired simultaneously from the top and bottom of a vertical cliff with elevation $\alpha = 30°$ and $\beta = 60°$ respectively and strikes the object simultaneously at the same point. If $a = 30\sqrt{3}$ m is the horizontal distance of the object from the cliff, then the height of the cliff is



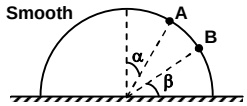
(A) 30 m (B) 45 m
(C) 60 m (D) 90 m

3. A projectile is required to hit a target whose co-ordinates relative to horizontal and vertical axes through the point of projection are (α, β). If the gun velocity is $\sqrt{2g\alpha}$. Then the condition when it is impossible to hit the target.

(A) $3\alpha > 4\beta$ (B) $4\beta > 3\alpha$
(C) $3\beta > 4\alpha$ (D) $4\alpha > 3\beta$

Space for rough work

4. A particle moves from rest at A on the surface of a smooth circular cylinder of radius r as shown. At B its leaves the cylinder. Then the equation relating α and β is



(A) $3\sin\alpha = 2\cos\beta$ (B) $2\sin\alpha = 3\cos\beta$
(C) $3\sin\beta = 2\cos\alpha$ (D) $2\sin\beta = 3\cos\alpha$

5. The gravitational field in a region is given by $E = \left(2\hat{i} + 3\hat{j}\right)$ N/kg. The equation of the line on which no work is done by the gravitational when the particle is moved
(A) $2y + 3x = 5$ (B) $5y + 2x = 3$
(C) $3y + 2x = 5$ (D) $3y + 5x = 2$

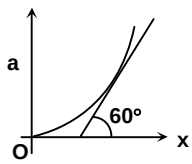
6. A rubber ball of mass m and radius r is submerged into water of density ρ to the depth h and released. Then to what height will the ball jump up above the surface of water?
(A) $\left(\rho\frac{4}{3}\pi r^3 - m\right)\frac{h}{m}$ (B) $\left(m - \rho\frac{4}{3}\pi r^3\right)\frac{h}{m}$
(C) $\left(\rho\frac{4}{3}\pi r^3 - m\right)\frac{h}{2m}$ (D) $\left(\rho\frac{4}{3}\pi r^3 + m\right)\frac{h}{m}$

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

7. In a YDSE set up the separation between the slits is 3×10^{-3} m. The distance between the slits and the screen is 1.5 m. Light of wavelength in the visible range (4000Å – 8000Å) is allowed to fall on the slits. The wave length in the visible region that will be present on the screen at a distance of 1.5×10^{-3} m from central maxima is
(A) 4285.7Å (B) 5000 Å
(C) 6000 Å (D) 7500Å

8. A particle starts moving with initial velocity 3 m/s along x-axis from origin. Its acceleration is varying with x in parabolic nature as shown in figure. At $x = \sqrt{3}$ m tangent to the graph makes an angle 60° with positive x-axis as shown in diagram. Then at $x = \sqrt{3}$

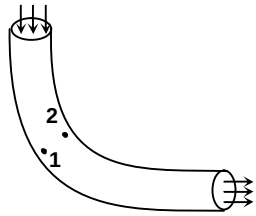


(A) $v = \sqrt{(\sqrt{3} + a)}$ m/s (B) $a = 1.5$ m/s^2
(C) $v = \sqrt{12}$ m/s (D) $a = 3$ m/s^2

Space for rough work

9. A vertical capillary tube with inside radius 0.25 mm is submerged into water so that the length of its part protruding over the water surface is equal to 25 mm. surface tension of water is 73×10^{-3} N/m and angle of contact is zero degree for glass and water, acceleration due to gravity is 9.8 m/s^2. Then choose correct statement.
(A) R = 0.25 mm (B) h = 59.6 mm
(C) R = 0.60 mm (D) h = 25 mm
where R is radius of meniscus and h is height of water in capillary tube.

10. Ideal fluid flows along a tube of uniform cross section, located in a horizontal plane and bent as shown in figure. The flow is steady, 1 and 2 are two points in the tube. If P_1 and P_2 are pressure at the two points and v_1 and v_2 are the respective velocities, then



(A) $P_1 < P_2$ (B) $P_1 > P_2$
(C) $v_1 < v_2$ (D) $v_1 > v_2$

Reasoning Type

This section contains **4 questions** numbered 11 to 14. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

11. STATEMENT-1: If the magnitude of momentum of a system of particle is zero, its kinetic energy may or may not be zero.
and
STATEMENT-2: the kinetic energy of a particle of mass m with momentum P is equal to $P^2/2m$
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

Space for rough work

12. STATEMENT-1: For an ideal fluid flowing through a horizontal pipe of non-uniform cross section. The pressure decreases with the decrease in cross-section.
and
STATEMENT-2: for ideal flow through a pipe of non-uniform cross section, the speed decreases with the increase in cross-section.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

13. STATEMENT-1: The electric field inside the cavity of a charged conductor is zero when there is no charge in the cavity.
and
STATEMENT-2: The flux of electric field through Gaussian surface inside the cavity with no charge is zero.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

14. STATEMENT-1: Light of frequency f illuminating a long narrow slit produces a diffraction pattern. On increasing the frequency slightly the pattern contracts toward the centre.
and
STATEMENT-2: For the same size obstacles, light waves of small wavelength undergo more diffraction.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

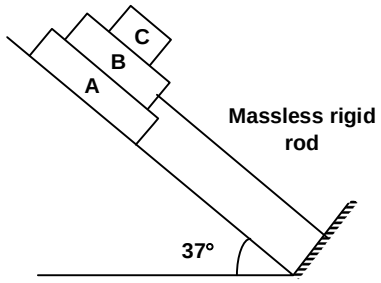
Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 15 to 17

In the shown figure m_A, m_B and m_C are the masses of the three blocks, the incline is frictionless m_A = 5 kg, m_B = 10 kg, m_C = 2kg. coefficient of friction between A and B is 0.5 and between B and C is 0.1 (Take g = 10 m/s^2)
On the basis of above data answer the following questions.



15. What is the frictional force on block A
(A) 30 N (B) 48 N
(C) 18 N (D) 78 N

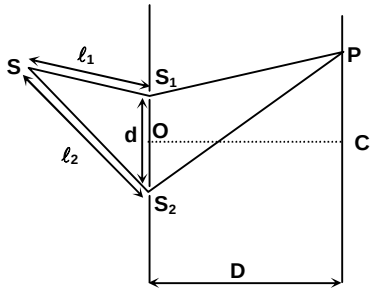
16. What is the frictional force on block C
(A) 12 N (B) 1.6 N
(C) 20 N (D) 13.6 N

17. What is the tension in the rod
(A) 91.6 N (B) 31.6 N
(C) 8.4 N (D) 90 N

Space for rough work

Paragraph for Question Nos. 18 to 20

In a young double slit experiment. The two slits are illuminated by a monochromatic light source S of wavelength λ = 500 nm. Distance between slits and screen is D = 2m and distance between two slits $S_1S_2 = 10$ μm. ℓ_1 = 20 μm and ℓ_2 = μm, as shown in the figure. Answer the following questions on the basis above data.



18. The value of θ relative to the central line OC, where maxima appear on the screen

(A) $\theta = \sin^{-1}\left[2\left(\frac{n}{40}-1\right)\right]$ (B) $\theta = \sin^{-1}\left[2\left(\frac{n}{20}-1\right)\right]$

(C) $\theta = \sin^{-1}\left[\left(\frac{n}{40}-1\right)\right]$ (D) $\theta = \sin^{-1}\left[\left(\frac{n}{20}-1\right)\right]$

19. How many maxima will appear on the screen

(A) 80 (B) 60

(C) 40 (D) 20

20. What should be minimum thickness of a slab of refractive index μ = 1.5 will be placed on the path of one ray so that minima occurs at C?

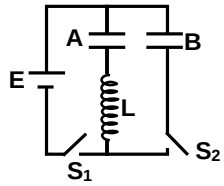
(A) 1000 nm (B) 250 nm

(C) 1500 nm (D) 500 nm

Space for rough work

Paragraph for Question Nos. 21 to 23

Two uncharged identical capacitors A and B, each of capacitance C, and an inductor of inductance L are arranged as shown in the adjacent figure. At t = 0, the switch S_1 is closed while switch S_2 remains open. At time $t = t_0 = \sqrt{LC}\,\frac{\pi}{2}$, switch S_2 is closed while switch S_1 is opened.



Read the above passage carefully and answer the following questions.

21. The charge on capacitor A after time t_0 is

(A) 2CE (B) $\frac{CE}{2}$

(C) CE (D) $\frac{CE}{4}$

22. The current flowing through the inductor at $t = t_0$ is

(A) $2CE\frac{1}{\sqrt{LC}}$ (B) 0

(C) $\frac{CE}{2}\times\frac{1}{\sqrt{LC}}$ (D) $CE\times\frac{1}{\sqrt{LC}}$

23. After switch S_2 is closed and S_1 is opened, the maximum value of current through the inductor is

(A) $\left(\sqrt{\frac{C}{L}}\right)E$ (B) $\left(\sqrt{\frac{C}{2L}}\right)E$

(C) $\left(\sqrt{\frac{3}{2}\frac{C}{L}}\right)E$ (D) $\left(\sqrt{\frac{5}{4}\frac{C}{L}}\right)E$

Space for rough work

Chemistry

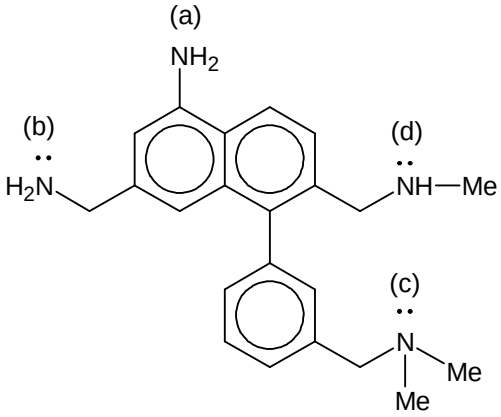
PART – II

SECTION – A

Straight Objective Type

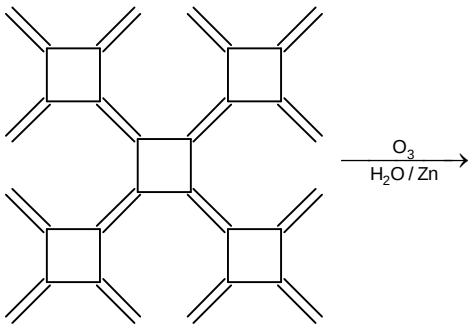
This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Which of the N atoms is least basic in the given compound?



(A) a (B) b
(C) c (D) d

2. How many formaldehyde will be formed in the given reaction?



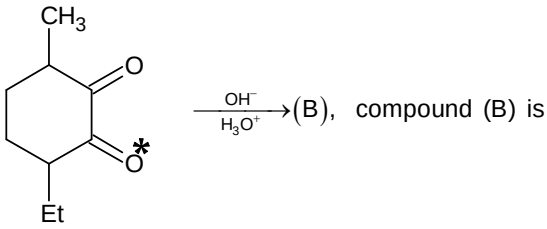
(A) 12 (B) 10
(C) 11 (D) 5

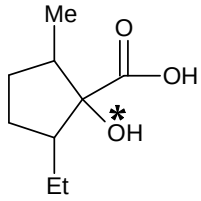
Space for rough work

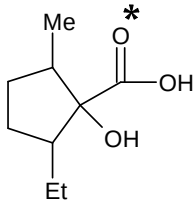
3. Select the incorrect statement about N_2F_4 and N_2H_4.
(i) In N_2F_4, d orbitals are contracted by electronegative fluorine atoms, but d-orbitals contraction is not possible by H-atoms in N_2H_4.
(ii) The N—N bond energy in N_2F_4 is more than N—N bond energy in N_2H_4
(iii) The N—N bond length in N_2F_4 is more than that of in N_2H_4
(iv) The N—N bond length in N_2F_4 is less than that of N_2H_4
Choose the correct code:
(A) (i), (ii), (iii) (B) (i) and (ii)
(C) (ii) and (iv) (D) (ii) and (iii)

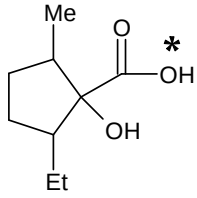
4. The radius of the Cl^- ion is 38% larger than that of the F^- ion but the radius of the Br^- ion is only 6.5% larger than that of the Cl^- ion. The relatively small difference in size between Cl^- and Br^- ion is due to the fact that
(A) The Br^- ion contains ten 3d electrons, which fails to shield the nuclear charge effectively.
(B) The Br^- ion contains ten 3d electrons, which shield the nuclear charge effectively.
(C) The Br^- ion contains six 4p electrons which shield the nuclear charge effectively.
(D) The Br^- ion contains ten 3d electrons and six 3p electrons, together they shield the nuclear charge effectively.

5. A cinema hall has equidistant rows 1m apart. The length of cinema hall is 287m and it has 287 rows. From one side of cinema hall, laughing gas N_2O is released and from the other side, weeping gas ($C_6H_5COCH_2Cl$) is released. In which rows, spectators will be laughing and weeping simultaneously?
(A) 100th row from weeping gas side (B) 100th row from laughing gas side
(C) 187th row from the weeping gas side (D) 185th row from the laughing gas side

6. 

(A) 

(B) 

(C) 

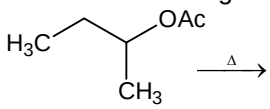
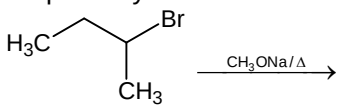
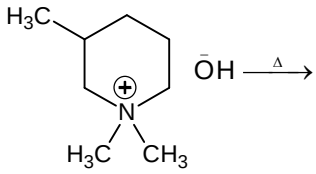
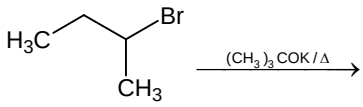
(D) All

Space for rough work

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

7. Which of the following statement(s) is/are correct?
   (A) For adiabatic expansion of an ideal gas, $TV^{\gamma-1} = \text{constant}$.
   (B) Work done in reversible isothermal expansion is greater than that done in reversible adiabatic expansion for the same increase of volume.
   (C) Buffer capacity is maximum when concentration of weak acid and salt of its conjugate base is equal.
   (D) Equilibrium constant of an exothermic reaction decreases with increase of temperature generally.

8. In which of the following reaction product formation takes place by Hofmann rule?
   (A) 
   (B) 
   (C) 
   (D) 

9. Select the correct statement(s):
   (A) Ionization energies of 5d elements are greater than those of 3d and 4d elements.
   (B) $[Fe(CN)_6]^{4-}$, $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ all are diamagnetic.
   (C) Holme's signal can be given by using (Ca_3P_2 + CaC_2).
   (D) Nitrolim is a mixture of calcium cyanide and carbon.

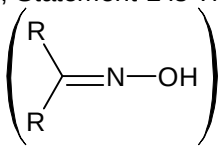
10. Nitrogen (I) oxide is produced by
   (A) thermal decomposition of ammonium nitrate
   (B) disproportionation of N_2O_4
   (C) thermal decomposition of ammonium nitrite
   (D) interaction of hydroxyl amine with nitrous acid

Space for rough work

Reasoning Type

This section contains **4 questions** numbered 11 to 14. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

11. STATEMENT-1: For π overlap the lobes of the atomic orbitals are perpendicular of the line joining the nuclei.
and
STATEMENT-2: In π molecular orbitals probability of finding electron is zero along the internuclear axis.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

12. STATEMENT-1: Oximes  are more acidic than hydroxylamine (NH_2OH).
and
STATEMENT-2: Conjugate base of oxime is resonance stablised.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Space for rough work

13. STATEMENT-1: In an atom, the velocity of electrons in the higher orbits keeps on decreasing.
and
STATEMENT-2: Velocity of electrons is inversely proportional to the radius of the orbit.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

14. STATEMENT-1: BrO_3^- shows two reactions as

$$SeO_3^{-\,-} + BrO_3^- + H^+ \longrightarrow SeO_4^- + Br_2 + H_2O$$

$$AsO_2^{-\,-} + BrO_3^- + H^+ \longrightarrow Br^- + AsO_4^- + H_2O$$

and

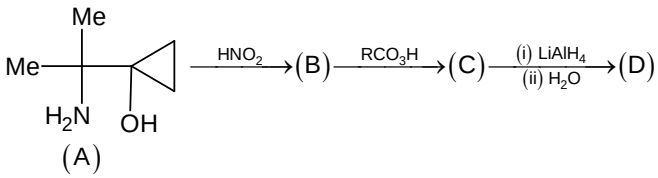
STATEMENT-2: The ratio of equivalent weights of BrO_3^- in two reaction 5/6
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Space for rough work

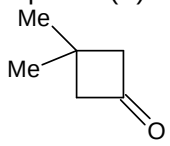
Comprehension Type

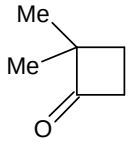
This section contains **3 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

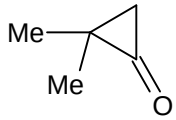
Paragraph for Question Nos. 15 to 17

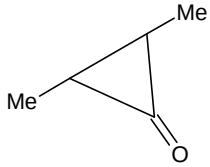


15. The compound (B) is

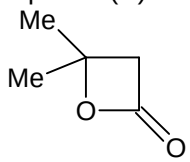
(A) 

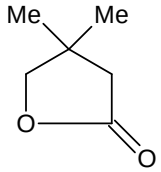
(B) 

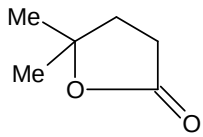
(C) 

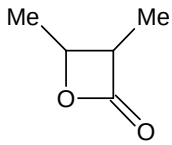
(D) 

16. The compound (C) is

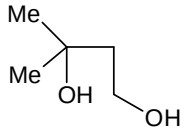
(A) 

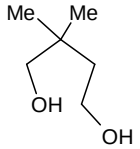
(B) 

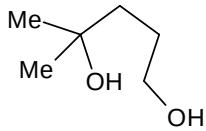
(C) 

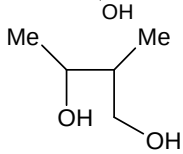
(D) 

17. The compound (D) is

(A) 

(B) 

(C) 

(D) 

Space for rough work

Paragraph for Question Nos. 18 to 20

$Pb(IO_3)_2$ is a sparingly soluble salt ($K_{sp} = 2.7 \times 10^{-13}$). To 35 ml 0.15 M $Pb(NO_3)_2$ solution, 15 ml of 0.8 M KIO_3 solution is added and a precipitate of $Pb(IO_3)_2$ is formed.

18. Which is a limiting reagent of the reaction that takes place in the solution?
(A) $Pb(IO_3)_2$ (B) $Pb(NO_3)_2$
(C) KIO_3 (D) Both B and C

19. What will be the molarity of IO_3^- ions in the solution after completion of the reaction.
(A) 0.152 (B) 0.081
(C) 0.41 (D) 0.03

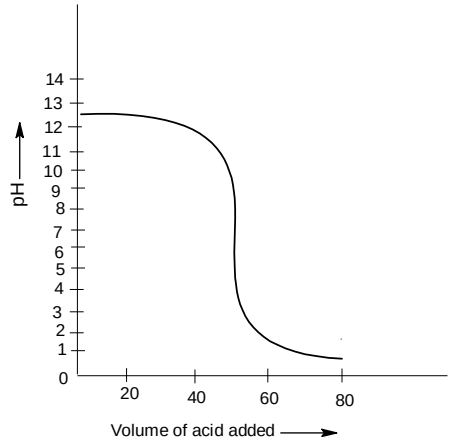
20. What will the molarity of Pb^{+2} ions in the solution after completion of the reaction
(A) 8.4×10^{-10} (B) 1.6×10^{-10}
(C) 3.0×10^{-10} (D) 6.1×10^{-10}

Space for rough work

Paragraph for Question Nos. 21 to 23

Read the following paragraph and answer the questions given below:

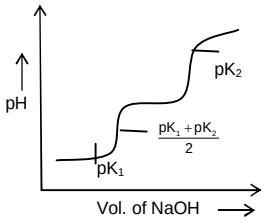
A titration curve graph of pH as a function of the volume of titrant, displays in detail how the pH changes over the course of an acid-base titration. Significantly, the pH changes most rapidly near the equivalence point. The exact shape of a titration curve depends on the relative strengths of acids and bases. Examine the titration curve below:

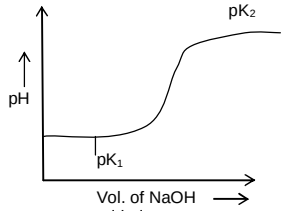


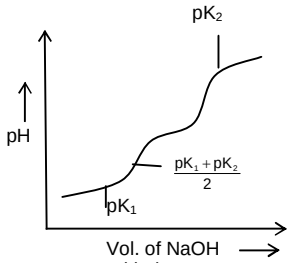
21. The suitable indicator for this titration will be
(A) bromomethyl blue (B) methyl orange
(C) methyl red (D) All of the above

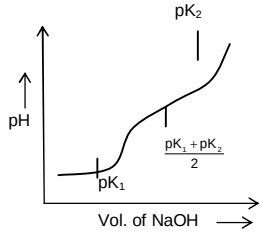
22. The pH at equivalence point is
(A) 11 (B) 7
(C) 3 (D) 2

23. Which of the following curve indicates the titration of a weak diprotic acid by NaOH of equivalent strength?

(A)


(B)


(C)


(D)


Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions** numbered 1 to 6. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If the parabola $y = ax^2 + bx + c$ has vertex at (4, 2) and $a \in [1, 3]$, then difference between the extreme values of abc is equal to,
   (A) 3600 (B) 144
   (C) 3456 (D) none of these

2. Let $f : R \to R$, $f(x) = \begin{cases} |x - [x]|, & [x] \text{ is odd} \\ |x - [x+1]|, & [x] \text{ is even} \end{cases}$, where [.] denotes greatest integer function, then $\int_{-2}^{4} f(x)dx$ is equal to
   (A) 5/2 (B) 3/2
   (C) 5 (D) 3

3. If $\lim_{x\to 0} \int_0^x \frac{t^2 dt}{(x - \sin x)\sqrt{a+t}} = 1$, then the value of a is
   (A) 4 (B) 2
   (C) 1 (D) none of these

4. Nine hundred distinct N-digit numbers are to be formed by using 6, 8 and 9 only. The smallest value of N for which this is possible, is
   (A) 6 (B) 7
   (C) 8 (D) 9

Space for rough work

5. If P, Q, R be midpoints of sides AB, BC and CA of ΔABC, then the ration of circumradius of ΔPQR to that of circumradius of ΔABC is
(A) 1 : 4 (B) 1 : 2
(C) 1 : 3 (D) none of these

6. The number of solutions of $\log_x 3 = 2x - 3$ is
(A) 3 (B) 1
(C) 2 (D) 0

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

7. $f(x) = \cos \pi(|x| + [x])$, then
(A) f is continuous at x = 1/2 (B) f is continuous at x = 0
(C) f is differentiable in (–1, 0) (D) f is differentiable in (0, 1)

8. Tangents are drawn to the circle $x^2 + y^2 = 50$ from a point 'P' lying on the x-axis. These tangents meet the y-axis at points 'P_1' and 'P_2'. Possible coordinates of 'P' so that area of triangle PP_1P_2 is minimum, is/are
(A) (10, 0) (B) $(10\sqrt{2}, 0)$
(C) (–10, 0) (D) $(-10\sqrt{2}, 0)$

9. Two particles start from the same point (2, –1), one moving 2 units along the line x + y = 1 and the other 5 units along x – 2y = 4. If the particles move towards increasing y, then their new position will be
(A) $(2 - \sqrt{2}, \sqrt{2} - 1)$ (B) $(2\sqrt{5} + 2, \sqrt{5} - 1)$
(C) $(2 + \sqrt{2}, \sqrt{2} + 1)$ (D) $(2\sqrt{5} - 2, \sqrt{5} - 1)$

10. If $\frac{x}{a} + \frac{y}{b} = 1$ is a tangent to the curve $x = 4t$, $y = \frac{4}{t}$, $t \in R$ then
(A) a > 0, b > 0 (B) a > 0, b < 0
(C) a < 0, b > 0 (D) a < 0, b < 0

Space for rough work

Reasoning Type

This section contains **4 questions** numbered 11 to 14. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

11. STATEMENT 1: $\pi^4 > 4^\pi$

STATEMENT 2: The function $y = x^x$ is decreasing $\forall\ x > \frac{1}{e}$

(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

12. If $P(\overline{B}) = \{P(A \cup B)\}^2$, then

STATEMENT 1: $P(A \cup B)$ is at least $\frac{\sqrt{5}-1}{2}$

STATEMENT 2: $A \cap (B \cap A) = A \cup (B \cap A)$
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

13. STATEMENT 1: $a + b + c + d + e = 8$ and $a^2 + b^2 + c^2 + d^2 + e^2 = 16$, where a, b, c, d, e are positive non-negative real numbers, then $e \in \left[0, \frac{16}{5}\right]$

STATEMENT 2: $\left(\frac{a+b+c+d}{4}\right)^2 \le \frac{a^2+b^2+c^2+d^2}{4}$
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

14. STATEMENT 1: Tangents at two distinct points of a cubic polynomial cannot coincide
STATEMENT 2: If P(x) is a polynomial of degree n ($n \ge 2$), then $P'(x) = k$ cannot hold for n or more distinct values of x
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1.
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1.
(C) Statement 1 is true and Statement 2 is false.
(D) Statement 1 is false and Statement 2 is true.

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 15 to 17

Read the following write up carefully and answer the following questions:
A circle C whose radius is 1 unit, touches the x–axis at point A. The centre Q of C lies in first quadrant. The tangent from origin O to the circle touches it at T and a point P lies on it such that ΔOAP is a right angled triangle at A and its perimeter is 8 units

15. The length of QP is
(A) $\frac{1}{2}$ (B) $\frac{4}{3}$
(C) $\frac{5}{3}$ (D) none of these

16. Equation of circle C is
(A) $\{x-(2+\sqrt{3})\}^2+(y-1)^2=1$ (B) $\{x-(\sqrt{3}+\sqrt{2})\}^2+(y-1)^2=1$
(C) $(x-\sqrt{3})^2+(y-2)^2=1$ (D) none of these

17. If tangent OT cuts the two parallel tangents (one of them is OA) at O and R, then equation of circle circumscribing the Δ ORQ is
(A) $x^2+y^2-\sqrt{3}x-y=0$ (B) $x^2+y^2-\sqrt{3}x-2y=0$
(C) $x^2+y^2-2\sqrt{3}x-2y=0$ (D) none of these

Paragraph for Question Nos. 18 to 20

Read the following write up carefully and answer the following questions:
The vertices of a ΔABC lies on a rectangular hyperbola such that the orthocentre of the triangle is (3, 2) and the asymptotes of the rectangular hyperbola are parallel to the coordinate axis. If the two perpendicular tangents of the hyperbola intersect at the point (1, 1)

18. The equation of the asymptotes is
(A) $xy-1=x-y$ (B) $xy+1=x+y$
(C) $2xy=x+y$ (D) none of these

19. Equation of the rectangular hyperbola is
(A) $xy=2x+y-2$ (B) $2xy=x+2y+5$
(C) $xy=x+y+1$ (D) none of these

20. Number of real normals that can drawn from the point (1, 1) to the rectangular hyperbola is
(A) 4 (B) 0
(C) 3 (D) 2

Space for rough work

Paragraph for Question Nos. 21 to 23

Read the following write up carefully and answer the following questions:
If $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ represents an ellipse, then
$h^2 < ab$ and $abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$. If for every point (x_1, y_1) satisfying above equation $(2h - x_1, 2k - y_1)$ also satisfy it, then (h, k) is centre of it. The length of semi major axis and minor axis is nothing but the maximum and minimum value of the distance of a point lying on the curve from its centre.

21. For the ellipse $2x^2 - 2xy + 4y^2 - (3 + \sqrt{2}) = 0$, the inclination of major axis of it with x-axis is

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{8}$

(C) $\frac{3\pi}{8}$ (D) $\frac{5\pi}{8}$

22. If the given ellipse is rotated so that it has its major axis coincident with x-axis, then its equation becomes

(A) $3(x^2 + y^2 - 1) = (y^2 - x^2 + 1)\sqrt{2}$ (B) $3(x^2 + y^2 - 1) = (y^2 - x^2 - 1)\sqrt{2}$

(C) $3(x^2 + y^2 - 1) = (y^2 - x^2 - 1)\sqrt{2}$ (D) $\sqrt{2}\ (x^2 - y^2 + 1) = 3(x^2 + y^2 - 1)$

23. The maximum area of the circle lying inside the given ellipse is

(A) π (B) $\frac{\pi}{2}$

(C) $\pi\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$ (D) $\pi\left(\frac{3-\sqrt{2}}{3+\sqrt{2}}\right)$

Space for rough work